

PIZZA



Pizza Sales Analysis

This project focuses on analyzing pizza sales data using SQL to uncover valuable insights into order patterns, customer preferences, sales performance, and revenue generation. The analysis involves writing SQL queries to retrieve and summarize key business metrics, including total orders, total revenue, the highest-priced pizza, the most popular pizza sizes, and the top-selling pizza types.

Using SQL techniques such as JOINS, aggregate functions, GROUP BY, subqueries, and window functions, the project explores pizza sales across different categories, hours of the day, and dates. Advanced analysis includes calculating revenue contribution percentages, tracking cumulative revenue over time, and identifying the top-performing pizza types within each category.

The goal of this project is to demonstrate practical SQL skills and transform raw sales data into meaningful business insights that can support data-driven decision-making.



Q. Retrieve the total number of orders placed.

```
select count(order_id) as Total_Order_Placed  
from orders;
```

Q. Calculate the total revenue generated from pizza sales

```
select round(sum(od.quantity * pizzas.price),2) as Total_Revenue  
from order_details as od  
join pizzas  
on od.pizza_id = pizzas.pizza_id;
```

Q. Identify the highest-priced pizza.

```
select pizza_types.name, pizzas.price as Highest_Priced_Pizza  
from pizzas join pizza_types  
on pizzas.pizza_type_id = pizza_types.pizza_type_id  
order by price desc limit 1;
```

Q. Identify the most common pizza size ordered

```
select pizzas.size, count(order_details.order_details_id) as total_id_count
from pizzas join order_details
on pizzas.pizza_id = order_details.pizza_id
group by pizzas.size order by total_id_count desc;
```

Q. List the top 5 most ordered pizza types along with their quantities

```
select pizza_types.name, sum(order_details.quantity) as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity desc limit 5;
```

Q. Join the necessary tables to find the total quantity of each pizza category ordered.

```
select pizza_types.category, sum(order_details.quantity) as quantity
from pizza_types
join pizzas
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by category order by quantity desc;
```

Q. Determine the distribution of orders by hour of the day

```
select hour(order_time) as hour,  
count(order_id) as order_count  
from orders  
group by hour;
```

Q. Join relevant tables to find the category-wise distribution of pizzas.

```
select category, count(name)  
from pizza_types  
group by category;
```

Q. Group the orders by date and calculate the average number of pizzas ordered per day.

- ```
select round(avg(quantity),0) as avg_pizza_ordered_per_day from
(select orders.order_date, sum(order_details.quantity) as quantity
from orders join order_details
on orders.order_id = order_details.order_id
group by orders.order_date) as order_quantity ;
```

**Q. Determine the top 3 most ordered pizza types based on revenue**

```
select pizza_types.name,
sum(pizzas.price * order_details.quantity) as revenue
from pizzas join order_details
on pizzas.pizza_id = order_details.pizza_id
join pizza_types
on pizza_types.pizza_type_id = pizzas.pizza_type_id
group by pizza_types.name
order by revenue desc limit 3;
```